

**HORNSUND, Svalbard and Jan Mayen**

WMO: 010030

Lat: 77.000N

Lon: 15.500E

Elev: 12

StdP: 101.18

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

**Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 3             | -22.1      | -20.5 | -27.9                         | 0.3 | -21.4 | -25.9 | 0.4 | -19.8 | 19.0                  | -7.9 | 17.2 | -7.4 | 4.8                   | 80   | 0.996 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |     |      |     |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-----|------|-----|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%  |      | 2%  |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB  | MCWB | DB  | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e) | (f)  | (g) | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 7             | 3.0                    | 8.1             | 6.1  | 7.2 | 5.6  | 6.6 | 5.2  | 6.6                 | 7.5  | 6.0 | 6.8  | 5.5 | 6.3  | 7.0                  | 80   |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 6.0                             | 5.8 | 6.7  | 5.3 | 5.5 | 6.1  | 4.7 | 5.3 | 5.7  | 22.0          | 7.7  | 20.6 | 6.9  | 19.5 | 6.4  | 10.1              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values<br>of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|-------------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                             |      | n=10 years |      | n=20 years |      | n=50 years |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                   | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 16.6              | 14.8 | 13.2 | -23.0                      | 10.2 | 3.0                | 1.0 | -25.2                                                 | 10.9 | -26.9      | 11.5 | -28.6      | 12.1 | -30.8      | 12.8 |
|                   |      |      | -23.5                      | 8.0  | 2.9                | 0.9 | -25.6                                                 | 8.6  | -27.3      | 9.2  | -28.9      | 9.7  | -31.0      | 10.3 |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual      | Jan      | Feb      | Mar       | Apr      | May      | Jun     | Jul     | Aug     | Sep     | Oct      | Nov      | Dec      |
|------------------------------------------------------------------------|-------------|-------------|----------|----------|-----------|----------|----------|---------|---------|---------|---------|----------|----------|----------|
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | (d) -2.7    | (e) -7.6 | (f) -7.9 | (g) -10.4 | (h) -7.1 | (i) -1.9 | (j) 2.6 | (k) 4.8 | (l) 4.4 | (m) 2.2 | (n) -1.7 | (o) -4.0 | (p) -6.6 |
|                                                                        | DBStd       | 6.61        | 6.41     | 5.75     | 5.80      | 5.20     | 2.99     | 1.54    | 1.26    | 1.38    | 2.41    | 3.73     | 4.36     | 5.41     |
|                                                                        | HDD10.0     | 4651        | 547      | 501      | 631       | 513      | 368      | 222     | 160     | 174     | 235     | 364      | 420      | 516      |
|                                                                        | HDD18.3     | 7692        | 805      | 734      | 889       | 763      | 627      | 472     | 418     | 433     | 485     | 622      | 670      | 774      |
|                                                                        | CDD10.0     | 0           | 0        | 0        | 0         | 0        | 0        | 0       | 0       | 0       | 0       | 0        | 0        | 0        |
|                                                                        | CDD18.3     | 0           | 0        | 0        | 0         | 0        | 0        | 0       | 0       | 0       | 0       | 0        | 0        | 0        |
|                                                                        | CDH23.3     | 0           | 0        | 0        | 0         | 0        | 0        | 0       | 0       | 0       | 0       | 0        | 0        | 0        |
|                                                                        | CDH26.7     | 0           | 0        | 0        | 0         | 0        | 0        | 0       | 0       | 0       | 0       | 0        | 0        | 0        |
| (14) Wind                                                              | WSAvg       | 5.6         | 6.7      | 7.2      | 6.3       | 6.0      | 5.5      | 4.1     | 4.5     | 4.3     | 5.3     | 5.1      | 6.4      | 6.6      |
| (15) Precipitation                                                     | PrecAvg     | 394         | 39       | 30       | 30        | 21       | 17       | 17      | 28      | 38      | 50      | 41       | 42       | 39       |
|                                                                        | PrecMax     | 736         | 96       | 81       | 58        | 64       | 47       | 70      | 92      | 86      | 142     | 141      | 168      | 114      |
|                                                                        | PrecMin     | 240         | 8        | 6        | 6         | 6        | 2        | 0       | 1       | 8       | 15      | 9        | 12       | 11       |
|                                                                        | PrecStd     | 95          | 22       | 20       | 15        | 14       | 9        | 16      | 22      | 22      | 32      | 29       | 32       | 21       |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 3.1      | 3.0      | 2.1       | 3.1      | 3.7      | 7.3     | 10.2    | 9.3     | 7.5     | 6.1      | 4.4      | 3.5      |
|                                                                        |             | MCWB        | 2.2      | 2.2      | 1.1       | 2.0      | 2.2      | 4.8     | 7.3     | 6.8     | 6.4     | 5.3      | 3.3      | 2.4      |
|                                                                        | 2%          | DB          | 2.0      | 1.7      | 1.1       | 1.8      | 2.7      | 6.2     | 8.4     | 8.1     | 6.8     | 5.2      | 3.1      | 2.2      |
|                                                                        |             | MCWB        | 1.2      | 0.8      | 0.3       | 1.0      | 1.4      | 4.2     | 6.0     | 6.2     | 5.8     | 4.5      | 2.0      | 1.3      |
|                                                                        | 5%          | DB          | 1.1      | 0.9      | 0.1       | 0.9      | 2.1      | 5.5     | 7.5     | 7.2     | 6.1     | 4.4      | 2.3      | 1.4      |
|                                                                        |             | MCWB        | 0.3      | 0.0      | -0.6      | 0.0      | 0.8      | 3.8     | 5.6     | 5.6     | 5.2     | 3.7      | 1.1      | 0.5      |
|                                                                        | 10%         | DB          | 0.2      | -0.2     | -1.5      | 0.0      | 1.5      | 4.7     | 6.7     | 6.4     | 5.5     | 3.5      | 1.5      | 0.5      |
|                                                                        |             | MCWB        | -0.7     | -1.1     | -2.5      | -0.9     | 0.3      | 3.3     | 5.2     | 5.1     | 4.6     | 2.6      | 0.4      | -0.3     |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 2.4      | 2.4      | 1.3       | 2.2      | 2.9      | 5.7     | 7.6     | 7.6     | 6.9     | 5.7      | 3.6      | 2.7      |
|                                                                        |             | MCDB        | 2.9      | 2.9      | 1.8       | 2.7      | 3.4      | 6.7     | 9.6     | 8.7     | 7.2     | 5.8      | 4.1      | 3.1      |
|                                                                        | 2%          | WB          | 1.4      | 1.2      | 0.5       | 1.1      | 1.8      | 4.7     | 6.6     | 6.6     | 6.1     | 4.7      | 2.2      | 1.5      |
|                                                                        |             | MCDB        | 1.8      | 1.6      | 1.0       | 1.8      | 2.4      | 5.7     | 7.7     | 7.6     | 6.5     | 5.0      | 2.9      | 2.1      |
|                                                                        | 5%          | WB          | 0.5      | 0.1      | -0.5      | 0.1      | 1.1      | 4.1     | 5.9     | 5.9     | 5.4     | 3.7      | 1.3      | 0.6      |
|                                                                        |             | MCDB        | 1.0      | 0.8      | 0.2       | 0.8      | 1.8      | 5.1     | 6.9     | 6.9     | 6.0     | 4.3      | 2.1      | 1.3      |
|                                                                        | 10%         | WB          | -0.5     | -1.1     | -2.4      | -0.8     | 0.5      | 3.5     | 5.4     | 5.2     | 4.7     | 2.7      | 0.6      | -0.2     |
|                                                                        |             | MCDB        | 0.1      | -0.2     | -1.6      | -0.1     | 1.3      | 4.6     | 6.4     | 6.2     | 5.4     | 3.4      | 1.4      | 0.5      |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 4.2      | 4.2      | 4.7       | 4.2      | 2.8      | 2.5     | 3.0     | 2.9     | 2.7     | 3.2      | 3.7      | 4.3      |
|                                                                        |             | MCDBR       | 3.9      | 4.0      | 5.5       | 3.9      | 3.0      | 3.6     | 4.2     | 3.9     | 3.0     | 3.3      | 3.4      | 4.0      |
|                                                                        | 5% WB       | MCWBR       | 4.0      | 3.8      | 5.5       | 3.6      | 2.5      | 2.6     | 3.0     | 2.9     | 2.8     | 3.2      | 3.2      | 3.9      |
|                                                                        |             | MCDWR       | 4.2      | 4.1      | 5.5       | 4.4      | 2.7      | 3.3     | 3.8     | 3.7     | 3.1     | 3.3      | 3.5      | 4.3      |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | N/A      | 0.224    | 0.275     | 0.297    | 0.312    | 0.302   | 0.306   | 0.297   | 0.261   | 0.367    | N/A      | N/A      |
|                                                                        |             | taud        | N/A      | 1.888    | 2.045     | 2.195    | 2.255    | 2.378   | 2.424   | 2.425   | 2.360   | 2.722    | N/A      | N/A      |
|                                                                        | Ebn at Noon | Ebn at Noon | 0        | 64       | 546       | 761      | 820      | 856     | 834     | 776     | 627     | 118      | 0        | 0        |
|                                                                        |             | Edh at Noon | 0        | 9        | 58        | 84       | 96       | 89      | 82      | 68      | 43      | 12       | 0        | 0        |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 0.00        | 0.04     | 0.71     | 2.35      | 3.83     | 4.73     | 3.75    | 2.51    | 0.95    | 0.13    | 0.00     | 0.00     | 0.00     |
|                                                                        | RadStd      | 0.00        | 0.01     | 0.10     | 0.34      | 0.30     | 0.42     | 0.36    | 0.26    | 0.11    | 0.02    | 0.00     | 0.00     | 0.00     |

**Historical Trends**

|      |                        | DBAvg   | Heating |         | Cooling |         |         | Degree-Days |         |         |         |
|------|------------------------|---------|---------|---------|---------|---------|---------|-------------|---------|---------|---------|
|      |                        |         | 99% DB  | 99% DP  | 1% DB   | 1% WB   | 1% DP   | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only           | (d) N/A | (e) N/A | (f) N/A | (g) N/A | (h) N/A | (i) N/A | (j) N/A     | (k) N/A | (l) N/A | (m) N/A |
| (47) | Regional (3 neighbors) | +1.52   | +4.63   | +4.10   | N/A     | N/A     | N/A     | -552        | -549    | N/A     | N/A     |